

JAMES VILLANUEVA

(310) 720-3956 | jvillan314@gmail.com | LinkedIn | GitHub

EDUCATION

University of California, San Diego

Bachelor of Science in Computer Science

GPA: 3.6

Sep. 2024 – June 2028

La Jolla, CA

- **Relevant Coursework:** Software Engineering, Data Structures and Object-Oriented Programming, Software Tools and Techniques, Discrete Mathematics, Computer Organization, Systems Programming, Algorithms
- Minor in Spanish Literature
- Member of ColorStack and Society of Hispanic Professional Engineers (SHPE)

EXPERIENCE

Software Engineering Team Lead

UCSD

Mar. 2026 – Jun. 2026

California

- Led a team of 11 through full-cycle design and development of a custom software application, coordinating architecture decisions, sprint priorities, and feature planning.
- Implemented application features, debugged issues during testing cycles, and collaborated with teammates to improve code quality and project consistency.
- Led frontend development, including UI design, wireframing, pull request reviews, and team communication to keep implementation aligned with project goals.

AI Conversation Simulation Research Assistant

FastAPI/Next.js Research Project

Research

UCSD

- Contributed to an AI-powered classroom discussion simulator with a FastAPI backend and Next.js frontend for modeling multi-agent conversations.
- Worked with generated personas, discussion topics, turn-taking logic, and saved transcripts/audio to study conversational structure and interaction patterns.
- Supported evaluation workflows for agent conversations, including turn-taking length, coherence, lexical diversity, stance detection, and persona consistency.

Lead Ambassador

Carrot Group

Feb. 2025 – Jun. 2026

California

- Represented the brand on college campuses through student outreach, clear communication, and relationship building.
- Collected student feedback and shared insights to improve future engagement strategies and event promotion.

PROJECTS

Club Connector | *TypeScript, React, Node.js, Next.js*

January 2026 – Present

- Developing a full-stack web application to help students discover and join campus organizations through a searchable, student-focused interface.
- Implementing dynamic frontend components using Next.js and React while coordinating with teammates through Git-based workflows and code reviews.

WatchTower | *JavaScript, Node.js, Supabase/Postgres, Clerk, Jest, Playwright*

April 2026 – June 2026

- Engineered an observability tool in an 11-person team using Agile methodologies, Git-based version control, and structured code review processes.
- Built a full-stack platform with a browser SDK, authenticated dashboard, backend event ingestion, and database-backed telemetry storage for application monitoring.

TECHNICAL SKILLS

Languages: Java, C/C++, TypeScript, JavaScript, HTML/CSS, ARM assembly, LaTeX, Python

Frameworks: React, Next.js, Node.js, FastAPI

Developer Tools: Git, GitHub, VS Code, Vim, Bash, Supabase/Postgres, Clerk, Playwright, Jest, OpenAI API

Professional Skills: Team leadership, code review, technical communication, debugging, collaboration, project planning